The Postulates of Quantum Mechanics (Cohen–Tannoudji, Ch. III) Section C. The physical implications of the Schrödinger equation

Beatriz Gonzalez del Rio

Outline

- A. Statement of the postulates
- B. The physical interpretation of the postulates concerning observables and their measurements
- C. The physical implications of the Schrödinger equation

Physical implications of the Schrödinger equation

The Schrödinger equation governs the **time evolution** of a quantum system:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle .$$

Physical implications of the Schrödinger equation

The Schrödinger equation governs the **time evolution** of a quantum system:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle .$$

It is a **first-order equation in time**.

Physical implications of the Schrödinger equation

The Schrödinger equation governs the **time evolution** of a quantum system:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle .$$

It is a **first-order equation in time**.
Given the initial state $|\psi(t_0)\rangle$, the state at any later time t is uniquely determined.

Physical implications of the Schrödinger equation

The Schrödinger equation governs the **time evolution** of a quantum system:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle .$$

It is a **first-order equation in time**.
Given the initial state $|\psi(t_0)\rangle$, the state at any later time t is uniquely determined.

There is no indeterminacy in the time evolution.

Physical implications of the Schrödinger equation

The Schrödinger equation governs the **time evolution** of a quantum system:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle .$$

It is a **first-order equation in time**.
Given the initial state $|\psi(t_0)\rangle$, the state at any later time t is uniquely determined.

There is no indeterminacy in the time evolution.

Indeterminacy appears only during a **measurement**.

Superposition principle

The Schrödinger equation is **linear and homogeneous**.

Superposition principle

The Schrödinger equation is **linear and homogeneous**.
If $|\psi_1(t)\rangle$ and $|\psi_2(t)\rangle$ are solutions, then:

$$\lambda_1 |\psi_1(t)\rangle + \lambda_2 |\psi_2(t)\rangle$$

is also a solution.

Superposition principle

The Schrödinger equation is **linear and homogeneous**.
If $|\psi_1(t)\rangle$ and $|\psi_2(t)\rangle$ are solutions, then:

$$\lambda_1 |\psi_1(t)\rangle + \lambda_2 |\psi_2(t)\rangle$$

is also a solution.
Time evolution is described by a linear operator:

$$|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle .$$

Superposition principle

The Schrödinger equation is **linear and homogeneous**.
If $|\psi_1(t)\rangle$ and $|\psi_2(t)\rangle$ are solutions, then:

$$\lambda_1 |\psi_1(t)\rangle + \lambda_2 |\psi_2(t)\rangle$$

is also a solution.
Time evolution is described by a linear operator:

$$|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle .$$

Quantum evolution preserves superpositions.

Conservation of probability (global)

Since $H(t)$ is Hermitian, the norm of the state vector remains constant.

Conservation of probability (global)

Since $H(t)$ is Hermitian, the norm of the state vector remains constant. Compute:

$$\frac{d}{dt}\langle\psi(t)|\psi(t)\rangle\,.$$

Conservation of probability (global)

Since $H(t)$ is Hermitian, the norm of the state vector remains constant. Compute:

$$\frac{d}{dt}\langle\psi(t)|\psi(t)\rangle\,.$$

Using:

$$\frac{d}{dt}|\psi(t)\rangle = \frac{1}{i\hbar}H(t)\,|\psi(t)\rangle\,,$$

Conservation of probability (global)

Since $H(t)$ is Hermitian, the norm of the state vector remains constant. Compute:

$$\frac{d}{dt}\langle\psi(t)|\psi(t)\rangle\,.$$

Using:

$$\frac{d}{dt}|\psi(t)\rangle = \frac{1}{i\hbar}H(t)\,|\psi(t)\rangle\,,$$

and its Hermitian conjugate,

$$\frac{d}{dt}\langle\psi(t)| = -\frac{1}{i\hbar}\langle\psi(t)|\,H^{\dagger}(t) = -\frac{1}{i\hbar}\langle\psi(t)|\,H(t),$$

Conservation of probability (global)

we obtain:

$$\frac{d}{dt}\langle\psi(t)|\psi(t)\rangle = -\frac{1}{i\hbar}\langle\psi(t)|\,H(t)\,|\psi(t)\rangle + \frac{1}{i\hbar}\langle\psi(t)|\,H(t)\,|\psi(t)\rangle = 0.$$

Conservation of probability (global)

we obtain:

$$\frac{d}{dt}\langle\psi(t)|\psi(t)\rangle = -\frac{1}{i\hbar}\langle\psi(t)|\,H(t)\,|\psi(t)\rangle + \frac{1}{i\hbar}\langle\psi(t)|\,H(t)\,|\psi(t)\rangle = 0.$$

If the state is normalized at t_0:

$$\langle\psi(t_0)|\psi(t_0)\rangle = \int \mathrm{d}^3 r|\psi(\mathbf{r}, t_0)|^2 = 1,$$

Conservation of probability (global)

we obtain:

$$\frac{d}{dt}\langle\psi(t)|\psi(t)\rangle = -\frac{1}{i\hbar}\langle\psi(t)|\,H(t)\,|\psi(t)\rangle + \frac{1}{i\hbar}\langle\psi(t)|\,H(t)\,|\psi(t)\rangle = 0.$$

If the state is normalized at t_0:

$$\langle\psi(t_0)|\psi(t_0)\rangle = \int \mathrm{d}^3 r|\psi(\mathbf{r},t_0)|^2 = 1,$$

$$\boxed{\langle\psi(t)|\psi(t)\rangle = \int \mathrm{d}^3 r|\psi(\mathbf{r},t)|^2 = \langle\psi(t_0)|\psi(t_0)\rangle = 1.}$$

Conservation of probability (global)

we obtain:

$$\frac{d}{dt}\langle\psi(t)|\psi(t)\rangle = -\frac{1}{i\hbar}\langle\psi(t)|\,H(t)\,|\psi(t)\rangle + \frac{1}{i\hbar}\langle\psi(t)|\,H(t)\,|\psi(t)\rangle = 0.$$

If the state is normalized at t_0:

$$\langle\psi(t_0)|\psi(t_0)\rangle = \int \mathrm{d}^3 r|\psi(\mathbf{r},t_0)|^2 = 1,$$

$$\boxed{\langle\psi(t)|\psi(t)\rangle = \int \mathrm{d}^3 r|\psi(\mathbf{r},t)|^2 = \langle\psi(t_0)|\psi(t_0)\rangle = 1.}$$

Time evolution preserves the **global probability**.

Local conservation of probability

Global conservation does not imply that the probability density, $\rho(\mathbf{r}, t)$, is constant at each point.

Local conservation of probability

Global conservation does not imply that the probability density, $\rho(\mathbf{r}, t)$, is constant at each point.

By analogy with electromagnetism, local conservation takes the form:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.$$

Local conservation of probability

Global conservation does not imply that the probability density, $\rho(\mathbf{r}, t)$, is constant at each point.
By analogy with electromagnetism, local conservation takes the form:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.$$

This is the **continuity equation**: if the probability of finding the particle in the (fixed) volume $\mathrm{d}^3 r$ about $\mathbf{r}$ varies over time, it means that the probability current has a non-zero flux across the surface which limits this volume element.

Local conservation of probability

Global conservation does not imply that the probability density, $\rho(\mathbf{r}, t)$, is constant at each point.
By analogy with electromagnetism, local conservation takes the form:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.$$

This is the **continuity equation**: if the probability of finding the particle in the (fixed) volume $\mathrm{d}^3 r$ about $\mathbf{r}$ varies over time, it means that the probability current has a non-zero flux across the surface which limits this volume element.

Probability behaves like a conserved fluid.

Probability current

For a particle in a scalar potential:

$$H = \frac{\mathbf{P}^2}{2m} + V(\mathbf{r}, t),$$

Start from the Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + V(\mathbf{r}, t)\psi.$$

Probability current

For a particle in a scalar potential:

$$H = \frac{\mathbf{P}^2}{2m} + V(\mathbf{r}, t),$$

Start from the Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + V(\mathbf{r}, t)\psi.$$

Take the complex conjugate:

$$-i\hbar \frac{\partial \psi^*}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi^* + V(\mathbf{r}, t)\psi^*,$$

(using that $V(\mathbf{r}, t)$ is real).

Probability current

Multiply the first by ψ^* and the conjugate by ψ:

$$i\hbar\,\psi^*\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\psi^*\nabla^2\psi + V|\psi|^2,$$

$$-i\hbar\,\psi\frac{\partial\psi^*}{\partial t} = -\frac{\hbar^2}{2m}\psi\nabla^2\psi^* + V|\psi|^2.$$

Probability current

Multiply the first by ψ^* and the conjugate by ψ:

$$i\hbar\,\psi^*\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\psi^*\nabla^2\psi + V|\psi|^2,$$

$$-i\hbar\,\psi\frac{\partial\psi^*}{\partial t} = -\frac{\hbar^2}{2m}\psi\nabla^2\psi^* + V|\psi|^2.$$

Subtract the second equation from the first (potentials cancel):

$$i\hbar\frac{\partial}{\partial t}\left(\psi^*\psi\right) = -\frac{\hbar^2}{2m}\left(\psi^*\nabla^2\psi - \psi\nabla^2\psi^*\right).$$

Probability current

Recognizing the probability density $\rho = |\psi|^2$:

$$\frac{\partial \rho}{\partial t} + \frac{\hbar}{2mi}\left(\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*\right) = 0.$$

Probability current

Recognizing the probability density $\rho = |\psi|^2$:

$$\frac{\partial \rho}{\partial t} + \frac{\hbar}{2mi}\left(\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*\right) = 0.$$

Setting the probability current, $\mathbf{J}(\mathbf{r}, t)$, as:

$$\mathbf{J}(\mathbf{r}, t) = \frac{\hbar}{2mi}\left(\psi^* \nabla \psi - \psi \nabla \psi^*\right)$$

Probability current

Recognizing the probability density $\rho = |\psi|^2$:

$$\frac{\partial \rho}{\partial t} + \frac{\hbar}{2mi}\left(\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*\right) = 0.$$

Setting the probability current, $\mathbf{J}(\mathbf{r}, t)$, as:

$$\mathbf{J}(\mathbf{r}, t) = \frac{\hbar}{2mi}\left(\psi^* \nabla \psi - \psi \nabla \psi^*\right)$$

the continuity equation is recovered:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0$$

Time evolution of the mean value of an observable

Let A be an observable. If the state $|\psi(t)\rangle$ is normalized, its mean value is:

$$\langle A \rangle(t) = \langle \psi(t) | \, A(t) \, | \psi(t) \rangle \, .$$

Time evolution of the mean value of an observable

Let A be an observable. If the state $|\psi(t)\rangle$ is normalized, its mean value is:

$$\langle A\rangle(t) = \langle\psi(t)|\, A(t)\, |\psi(t)\rangle\,.$$

Since $|\psi(t)\rangle$ evolves in time, $\langle A\rangle(t)$ is, in general, a function of t.

Time evolution of the mean value of an observable

Let A be an observable. If the state $|\psi(t)\rangle$ is normalized, its mean value is:

$$\langle A\rangle(t) = \langle\psi(t)|\, A(t)\, |\psi(t)\rangle\,.$$

Since $|\psi(t)\rangle$ evolves in time, $\langle A\rangle(t)$ is, in general, a function of t.
We now compute:

$$\frac{d}{dt}\langle A\rangle(t).$$

General formula for $\frac{d}{dt}\langle A \rangle$

Differentiate:

$$\frac{d}{dt}\langle A \rangle = \frac{d}{dt}\left\langle \psi(t) \right| A(t) \left| \psi(t) \right\rangle .$$

General formula for $\frac{d}{dt}\langle A \rangle$

Differentiate:

$$\frac{d}{dt}\langle A \rangle = \frac{d}{dt}\left\langle \psi(t) \right| A(t) \left| \psi(t) \right\rangle .$$

Using the product rule:

$$\frac{d}{dt}\langle A \rangle = \left[\frac{d}{dt}\left\langle \psi \right|\right] A \left| \psi \right\rangle + \left\langle \psi \right| A \left[\frac{d}{dt}\left| \psi \right\rangle\right] + \left\langle \psi \right| \frac{\partial A}{\partial t} \left| \psi \right\rangle .$$

General formula for $\frac{d}{dt}\langle A \rangle$

Differentiate:

$$\frac{d}{dt}\langle A \rangle = \frac{d}{dt}\left\langle \psi(t) \right| A(t) \left| \psi(t) \right\rangle .$$

Using the product rule:

$$\frac{d}{dt}\langle A \rangle = \left[\frac{d}{dt}\left\langle \psi \right|\right] A \left| \psi \right\rangle + \left\langle \psi \right| A \left[\frac{d}{dt}\left| \psi \right\rangle\right] + \left\langle \psi \right| \frac{\partial A}{\partial t} \left| \psi \right\rangle .$$

Using the Schrödinger equation, we obtain:

$$\boxed{\frac{d}{dt}\langle A \rangle = \frac{1}{i\hbar}\langle [A, H] \rangle + \left\langle \frac{\partial A}{\partial t} \right\rangle}$$

Relation with classical mechanics

The quantity $\langle A \rangle(t)$ is a number depending only on time.

Relation with classical mechanics

The quantity $\langle A \rangle(t)$ is a number depending only on time.
The Hermitian operator $A(\mathbf{R}, \mathbf{P}, t)$ obtained using the quantization rules on the classical quantity $A(\mathbf{r}, \mathbf{p}, t)$, does **not** have a dependence on time in $\mathbf{R}$ and $\mathbf{P}$, but through the quantum state vector $|\psi(t)\rangle$.

Relation with classical mechanics

The quantity $\langle A \rangle(t)$ is a number depending only on time.
The Hermitian operator $A(\mathbf{R}, \mathbf{P}, t)$ obtained using the quantization rules on the classical quantity $A(\mathbf{r}, \mathbf{p}, t)$, does **not** have a dependence on time in $\mathbf{R}$ and $\mathbf{P}$, but through the quantum state vector $|\psi(t)\rangle$.
In the position representation:

$$\langle A \rangle = \int d^3 r \; \psi^*(\mathbf{r}, t) \, A\left(\mathbf{r}, \frac{\hbar}{i}\nabla, t\right) \psi(\mathbf{r}, t).$$

Relation with classical mechanics

The quantity $\langle A \rangle(t)$ is a number depending only on time.
The Hermitian operator $A(\mathbf{R}, \mathbf{P}, t)$ obtained using the quantization rules on the classical quantity $A(\mathbf{r}, \mathbf{p}, t)$, does **not** have a dependence on time in $\mathbf{R}$ and $\mathbf{P}$, but through the quantum state vector $|\psi(t)\rangle$.
In the position representation:

$$\langle A \rangle = \int d^3r \; \psi^*(\mathbf{r}, t) \, A\left(\mathbf{r}, \frac{\hbar}{i}\nabla, t\right) \psi(\mathbf{r}, t).$$

Integration over $\mathbf{r}$ leads to a number which only depends on t.

Application to R and P (Ehrenfest's theorem)

Consider a spinless particle in a scalar stationary potential:

$$H = \frac{\mathbf{P}^2}{2m} + V(\mathbf{R}).$$

Application to R and P (Ehrenfest's theorem)

Consider a spinless particle in a scalar stationary potential:

$$H = \frac{\mathbf{P}^2}{2m} + V(\mathbf{R}).$$

From the general formula of $\frac{d}{dt}\langle A \rangle$, we obtain:

$$\frac{d}{dt}\langle \mathbf{R} \rangle = \frac{1}{i\hbar}\langle [\mathbf{R}, H] \rangle ,$$

$$\frac{d}{dt}\langle \mathbf{P} \rangle = \frac{1}{i\hbar}\langle [\mathbf{P}, H] \rangle .$$

Commutator evaluation

Using the commutation relations:

$$[\mathbf{R}, \mathbf{P}^2] = 2i\hbar\,\mathbf{P}, \qquad [\mathbf{P}, V(\mathbf{R})] = -i\hbar\,\nabla V(\mathbf{R})$$

Commutator evaluation

Using the commutation relations:

$$[\mathbf{R}, \mathbf{P}^2] = 2i\hbar\,\mathbf{P}, \qquad [\mathbf{P}, V(\mathbf{R})] = -\,i\hbar\,\nabla V(\mathbf{R})$$

we find:

$$\boxed{\frac{d}{dt}\langle \mathbf{R} \rangle = \frac{1}{m}\langle \mathbf{P} \rangle}$$

$$\boxed{\frac{d}{dt}\langle \mathbf{P} \rangle = -\,\langle \nabla V(\mathbf{R}) \rangle}$$

Physical interpretation

Define:

$$\langle \mathbf{R} \rangle(t) = (\langle X \rangle, \langle Y \rangle, \langle Z \rangle).$$

Physical interpretation

Define:

$$\langle \mathbf{R} \rangle(t) = (\langle X \rangle, \langle Y \rangle, \langle Z \rangle).$$

This represents the **center of the wave packet**.

Physical interpretation

Define:

$$\langle \mathbf{R} \rangle(t) = (\langle X \rangle, \langle Y \rangle, \langle Z \rangle).$$

This represents the **center of the wave packet**.
From Ehrenfest:

$$m \frac{d^2}{dt^2} \langle \mathbf{R} \rangle = - \langle \nabla V(\mathbf{R}) \rangle.$$

Physical interpretation

Define:

$$\langle \mathbf{R} \rangle(t) = (\langle X \rangle, \langle Y \rangle, \langle Z \rangle).$$

This represents the **center of the wave packet**.
From Ehrenfest:

$$m \frac{d^2}{dt^2} \langle \mathbf{R} \rangle = - \langle \nabla V(\mathbf{R}) \rangle.$$

Does the center of the wave packet obey classical mechanics?

Comparison with classical force

Classically:

$$\mathbf{F}_{cl} = -\nabla V(\mathbf{r}).$$

Comparison with classical force

Classically:

$$\mathbf{F}_{cl} = -\nabla V(\mathbf{r}).$$

Quantum mechanically:

$$m\frac{d^2}{dt^2}\langle \mathbf{R} \rangle = -\langle \nabla V(\mathbf{R}) \rangle.$$

Comparison with classical force

Classically:

$$\mathbf{F}_{cl} = -\nabla V(\mathbf{r}).$$

Quantum mechanically:

$$m\frac{d^2}{dt^2}\langle \mathbf{R} \rangle = -\langle \nabla V(\mathbf{R}) \rangle.$$

In general:

$$\boxed{\langle \nabla V(\mathbf{R}) \rangle \neq \nabla V(\langle \mathbf{R} \rangle)}$$

Comparison with classical force

Classically:

$$\mathbf{F}_{cl} = -\nabla V(\mathbf{r}).$$

Quantum mechanically:

$$m\frac{d^2}{dt^2}\langle \mathbf{R} \rangle = -\langle \nabla V(\mathbf{R}) \rangle.$$

In general:

$$\boxed{\langle \nabla V(\mathbf{R}) \rangle \neq \nabla V(\langle \mathbf{R} \rangle)}$$

The mean of a function is not equal to the function evaluated at the mean.

Classical limit: localized wave packet

Suppose the wave packet is highly localized.

Classical limit: localized wave packet

Suppose the wave packet is highly localized.
If $V(\mathbf{r})$ varies slowly over the spatial extension of the packet,

Classical limit: localized wave packet

Suppose the wave packet is highly localized.
If $V(\mathbf{r})$ varies slowly over the spatial extension of the packet, then:

$$\langle \nabla V(\mathbf{R}) \rangle \approx \nabla V(\langle \mathbf{R} \rangle).$$

Classical limit: localized wave packet

Suppose the wave packet is highly localized.
If $V(\mathbf{r})$ varies slowly over the spatial extension of the packet, then:

$$\langle \nabla V(\mathbf{R}) \rangle \approx \nabla V(\langle \mathbf{R} \rangle).$$

Thus:

$$m \frac{d^2}{dt^2} \langle \mathbf{R} \rangle = - \nabla V(\langle \mathbf{R} \rangle).$$

Classical limit: localized wave packet

Suppose the wave packet is highly localized.
If $V(\mathbf{r})$ varies slowly over the spatial extension of the packet, then:

$$\langle \nabla V(\mathbf{R}) \rangle \approx \nabla V(\langle \mathbf{R} \rangle).$$

Thus:

$$m \frac{d^2}{dt^2} \langle \mathbf{R} \rangle = - \nabla V(\langle \mathbf{R} \rangle).$$

In this limiting case, quantum motion reproduces classical mechanics.

Conservative systems

When the Hamiltonian does **not depend explicitly on time**,

$$\frac{\partial H}{\partial t} = 0,$$

the system is said to be **conservative**.

Conservative systems

When the Hamiltonian does **not depend explicitly on time**,

$$\frac{\partial H}{\partial t} = 0,$$

the system is said to be **conservative**.
In classical mechanics, this implies:

Total energy is conserved.

Conservative systems

When the Hamiltonian does **not depend explicitly on time**,

$$\frac{\partial H}{\partial t} = 0,$$

the system is said to be **conservative**.
In classical mechanics, this implies:

Total energy is conserved.

In quantum mechanics, we shall see that conservative systems possess important special properties.

Eigenvalue equation of the Hamiltonian

Consider the eigenvalue equation:

$$H \left|\phi_{n,\tau}\right\rangle = E_n \left|\phi_{n,\tau}\right\rangle .$$

Eigenvalue equation of the Hamiltonian

Consider the eigenvalue equation:

$$H\,|\phi_{n,\tau}\rangle = E_n\,|\phi_{n,\tau}\rangle\,.$$

Assume for simplicity that the spectrum of H is discrete.

Eigenvalue equation of the Hamiltonian

Consider the eigenvalue equation:

$$H\,|\phi_{n,\tau}\rangle = E_n\,|\phi_{n,\tau}\rangle\,.$$

Assume for simplicity that the spectrum of H is discrete.
Since H does not depend explicitly on time:

$$E_n \text{ and } |\phi_{n,\tau}\rangle \text{ are time independent.}$$

Eigenvalue equation of the Hamiltonian

Consider the eigenvalue equation:

$$H\,|\phi_{n,\tau}\rangle = E_n\,|\phi_{n,\tau}\rangle\,.$$

Assume for simplicity that the spectrum of H is discrete.
Since H does not depend explicitly on time:

$$E_n \text{ and } |\phi_{n,\tau}\rangle \text{ are time independent.}$$

The eigenstates form a complete basis.

Expansion of the state

Any state can be expanded as:

$$|\psi(t)\rangle = \sum_{n,\tau} c_{n,\tau}(t)\ |\phi_{n,\tau}\rangle\ .$$

Expansion of the state

Any state can be expanded as:

$$|\psi(t)\rangle = \sum_{n,\tau} c_{n,\tau}(t)\ |\phi_{n,\tau}\rangle\,.$$

The coefficients are:

$$c_{n,\tau}(t) = \langle\phi_{n,\tau}|\psi(t)\rangle\,.$$

Expansion of the state

Any state can be expanded as:

$$|\psi(t)\rangle = \sum_{n,\tau} c_{n,\tau}(t)\ |\phi_{n,\tau}\rangle\,.$$

The coefficients are:

$$c_{n,\tau}(t) = \langle\phi_{n,\tau}|\psi(t)\rangle\,.$$

Projecting the Schrödinger equation:

$$i\hbar\frac{d}{dt}c_{n,\tau}(t) = E_n\, c_{n,\tau}(t).$$

Time evolution in a conservative system

The differential equation:

$$i\hbar \frac{d}{dt} c_{n,\tau}(t) = E_n \, c_{n,\tau}(t)$$

Time evolution in a conservative system

The differential equation:

$$i\hbar \frac{d}{dt} c_{n,\tau}(t) = E_n \, c_{n,\tau}(t)$$

has the solution:

$$c_{n,\tau}(t) = c_{n,\tau}(t_0) e^{-iE_n(t-t_0)/\hbar}.$$

Time evolution in a conservative system

The differential equation:

$$i\hbar \frac{d}{dt} c_{n,\tau}(t) = E_n \, c_{n,\tau}(t)$$

has the solution:

$$c_{n,\tau}(t) = c_{n,\tau}(t_0) e^{-iE_n(t-t_0)/\hbar}.$$

Thus:

$$\boxed{|\psi(t)\rangle = \sum_{n,\tau} c_{n,\tau}(t_0) e^{-iE_n(t-t_0)/\hbar} \, |\phi_{n,\tau}\rangle}$$

Stationary states

Suppose the system is initially in an eigenstate of H with eigenvalue E_n, possibly degenerate:

Stationary states

Suppose the system is initially in an eigenstate of H with eigenvalue E_n, possibly degenerate:

$$|\psi(t_0)\rangle = \sum_\tau c_{n,\tau}(t_0)\,|\phi_{n,\tau}\rangle\,,$$

Stationary states

Suppose the system is initially in an eigenstate of H with eigenvalue E_n, possibly degenerate:

$$|\psi(t_0)\rangle = \sum_\tau c_{n,\tau}(t_0)\,|\phi_{n,\tau}\rangle\,,$$

where all the states $|\phi_{n,\tau}\rangle$ share the same eigenvalue:

$$H\,|\phi_{n,\tau}\rangle = E_n\,|\phi_{n,\tau}\rangle\,.$$

Stationary states

Suppose the system is initially in an eigenstate of H with eigenvalue E_n, possibly degenerate:

$$|\psi(t_0)\rangle = \sum_\tau c_{n,\tau}(t_0)\,|\phi_{n,\tau}\rangle\,,$$

where all the states $|\phi_{n,\tau}\rangle$ share the same eigenvalue:

$$H\,|\phi_{n,\tau}\rangle = E_n\,|\phi_{n,\tau}\rangle\,.$$

Time evolution gives:

$$|\psi(t)\rangle = \sum_\tau c_{n,\tau}(t_0)e^{-iE_n(t-t_0)/\hbar}\,|\phi_{n,\tau}\rangle\,.$$

Stationary states

Factorizing the common phase:

$$\boxed{|\psi(t)\rangle = e^{-iE_n(t-t_0)/\hbar}\,|\psi(t_0)\rangle}$$

Stationary states

Factorizing the common phase:

$$\boxed{|\psi(t)\rangle = e^{-iE_n(t-t_0)/\hbar}\,|\psi(t_0)\rangle}$$

The state changes only by a **global phase factor**.

Stationary states

Factorizing the common phase:

$$\boxed{|\psi(t)\rangle = e^{-iE_n(t-t_0)/\hbar}\,|\psi(t_0)\rangle}$$

The state changes only by a **global phase factor**.
All physical properties remain constant in time.

Stationary states

Factorizing the common phase:

$$\boxed{|\psi(t)\rangle = e^{-iE_n(t-t_0)/\hbar}\,|\psi(t_0)\rangle}$$

The state changes only by a **global phase factor**.
All physical properties remain constant in time.
Eigenstates of H are called **stationary states**.

Conservation of energy

If the system is in an eigenstate of H, a measurement of energy gives:

$$E_n.$$

Conservation of energy

If the system is in an eigenstate of H, a measurement of energy gives:

$$E_n.$$

After the measurement, the system remains in that same eigenstate.

Conservation of energy

If the system is in an eigenstate of H, a measurement of energy gives:

$$E_n.$$

After the measurement, the system remains in that same eigenstate.
At any later time, a second measurement yields the same result:

$$E_n.$$

Conservation of energy

If the system is in an eigenstate of H, a measurement of energy gives:

$$E_n.$$

After the measurement, the system remains in that same eigenstate.
At any later time, a second measurement yields the same result:

$$E_n.$$

Energy is conserved in a conservative quantum system.

Constants of the motion

By definition, an observable A is a **constant of the motion** if:

Constants of the motion

By definition, an observable A is a **constant of the motion** if:

$$\frac{\partial A}{\partial t} = 0 \qquad \text{and} \qquad [A, H] = 0.$$

Constants of the motion

By definition, an observable A is a **constant of the motion** if:

$$\frac{\partial A}{\partial t} = 0 \qquad \text{and} \qquad [A, H] = 0.$$

Using the general formula:

$$\frac{d}{dt}\langle A \rangle = \frac{1}{i\hbar}\langle [A, H] \rangle + \left\langle \frac{\partial A}{\partial t} \right\rangle = 0,$$

Constants of the motion

By definition, an observable A is a **constant of the motion** if:

$$\frac{\partial A}{\partial t} = 0 \qquad \text{and} \qquad [A, H] = 0.$$

Using the general formula:

$$\frac{d}{dt}\langle A \rangle = \frac{1}{i\hbar}\langle [A, H] \rangle + \left\langle \frac{\partial A}{\partial t} \right\rangle = 0,$$

The mean value of A does not evolve in time.

Common eigenvectors

If A and H commute, they possess a common set of eigenvectors:

$$H\left|\phi_{n,p,\tau}\right\rangle = E_n\left|\phi_{n,p,\tau}\right\rangle,$$

$$A\left|\phi_{n,p,\tau}\right\rangle = a_p\left|\phi_{n,p,\tau}\right\rangle.$$

Common eigenvectors

If A and H commute, they possess a common set of eigenvectors:

$$H\left|\phi_{n,p,\tau}\right\rangle = E_n\left|\phi_{n,p,\tau}\right\rangle,$$

$$A\left|\phi_{n,p,\tau}\right\rangle = a_p\left|\phi_{n,p,\tau}\right\rangle.$$

Assume the spectra are discrete.

Common eigenvectors

If A and H commute, they possess a common set of eigenvectors:

$$H\left|\phi_{n,p,\tau}\right\rangle = E_n\left|\phi_{n,p,\tau}\right\rangle,$$

$$A\left|\phi_{n,p,\tau}\right\rangle = a_p\left|\phi_{n,p,\tau}\right\rangle.$$

Assume the spectra are discrete.
Since these states are eigenstates of H, they are **stationary states**.

Common eigenvectors

If A and H commute, they possess a common set of eigenvectors:

$$H\,|\phi_{n,p,\tau}\rangle = E_n\,|\phi_{n,p,\tau}\rangle\,,$$

$$A\,|\phi_{n,p,\tau}\rangle = a_p\,|\phi_{n,p,\tau}\rangle\,.$$

Assume the spectra are discrete.
Since these states are eigenstates of H, they are **stationary states**.
When A is a constant of the motion, the stationary states always remain, for all t, eigenstates of A with the same eigenvalues a_p, called **good quantum numbers**.

Probability conservation

Expand an arbitrary state:

$$|\psi(t_0)\rangle = \sum_{n,p,\tau} c_{n,p,\tau}(t_0)\,|\phi_{n,p,\tau}\rangle\,.$$

Probability conservation

Expand an arbitrary state:

$$|\psi(t_0)\rangle = \sum_{n,p,\tau} c_{n,p,\tau}(t_0)\,|\phi_{n,p,\tau}\rangle\,.$$

Time evolution:

$$c_{n,p,\tau}(t) = c_{n,p,\tau}(t_0)e^{-iE_n(t-t_0)/\hbar} \quad \text{with} \quad |c_{n,p,\tau}(t)|^2 = |c_{n,p,\tau}(t_0)|^2.$$

Probability conservation

Expand an arbitrary state:

$$|\psi(t_0)\rangle = \sum_{n,p,\tau} c_{n,p,\tau}(t_0)\,|\phi_{n,p,\tau}\rangle\,.$$

Time evolution:

$$c_{n,p,\tau}(t) = c_{n,p,\tau}(t_0)e^{-iE_n(t-t_0)/\hbar} \quad \text{with} \quad |c_{n,p,\tau}(t)|^2 = |c_{n,p,\tau}(t_0)|^2.$$

Therefore the probability

$$\mathcal{P}(a_p, t) = \sum_{n,\tau} |c_{n,p,\tau}(t)|^2 = \mathcal{P}(a_p, t_0)$$

is independent of time.

Time evolution of a general observable

Let B be an observable (not necessarily commuting with H).

Time evolution of a general observable

Let B be an observable (not necessarily commuting with H).
For a conservative system:

$$|\psi(t)\rangle = \sum_{n,\tau} c_{n,\tau}(t_0) e^{-iE_n(t-t_0)/\hbar} |\phi_{n,\tau}\rangle \, .$$

Time evolution of a general observable

Let B be an observable (not necessarily commuting with H).
For a conservative system:

$$|\psi(t)\rangle = \sum_{n,\tau} c_{n,\tau}(t_0) e^{-iE_n(t-t_0)/\hbar} |\phi_{n,\tau}\rangle \, .$$

We can compute explicitly:

$$\langle B \rangle(t) = \langle \psi(t) | B | \psi(t) \rangle .$$

Bohr frequencies

Inserting the expansion, we obtain:

Bohr frequencies

Inserting the expansion, we obtain:

$$\langle B \rangle(t) = \sum_{n,n',\tau,\tau'} c^*_{n',\tau'}(t_0) c_{n,\tau}(t_0) \langle \phi_{n',\tau'} | B | \phi_{n,\tau} \rangle e^{i(E_{n'} - E_n)(t-t_0)/\hbar}.$$

Bohr frequencies

Inserting the expansion, we obtain:

$$\langle B \rangle(t) = \sum_{n,n',\tau,\tau'} c^*_{n',\tau'}(t_0) c_{n,\tau}(t_0) \langle \phi_{n',\tau'} | B | \phi_{n,\tau} \rangle e^{i(E_{n'} - E_n)(t-t_0)/\hbar}.$$

The time dependence appears through:

$$\omega_{n'n} = \frac{E_{n'} - E_n}{\hbar} = 2\pi\nu_{n'n}.$$

Bohr frequencies

Inserting the expansion, we obtain:

$$\langle B \rangle(t) = \sum_{n,n',\tau,\tau'} c^*_{n',\tau'}(t_0) c_{n,\tau}(t_0) \langle \phi_{n',\tau'} | B | \phi_{n,\tau} \rangle e^{i(E_{n'} - E_n)(t-t_0)/\hbar}.$$

The time dependence appears through:

$$\omega_{n'n} = \frac{E_{n'} - E_n}{\hbar} = 2\pi\nu_{n'n}.$$

These frequencies are called **Bohr frequencies**.

Bohr frequencies

Inserting the expansion, we obtain:

$$\langle B \rangle(t) = \sum_{n,n',\tau,\tau'} c^*_{n',\tau'}(t_0) c_{n,\tau}(t_0) \langle \phi_{n',\tau'} | B | \phi_{n,\tau} \rangle e^{i(E_{n'} - E_n)(t-t_0)/\hbar}.$$

The time dependence appears through:

$$\omega_{n'n} = \frac{E_{n'} - E_n}{\hbar} = 2\pi\nu_{n'n}.$$

These frequencies are called **Bohr frequencies**.
They depend only on the energy differences, not on B.

Selection rules

The weights of the oscillating terms depend on:

$$\langle \phi_{n',\tau'} | B | \phi_{n,\tau} \rangle .$$

Selection rules

The weights of the oscillating terms depend on:

$$\langle \phi_{n',\tau'} | B | \phi_{n,\tau} \rangle.$$

If certain matrix elements vanish, the corresponding frequencies are absent.

Selection rules

The weights of the oscillating terms depend on:

$$\langle \phi_{n',\tau'} | B | \phi_{n,\tau} \rangle .$$

If certain matrix elements vanish, the corresponding frequencies are absent.

Allowed transitions are determined by $\langle n' | B | n \rangle \neq 0.$

Selection rules

The weights of the oscillating terms depend on:

$$\langle \phi_{n',\tau'} | B | \phi_{n,\tau} \rangle .$$

If certain matrix elements vanish, the corresponding frequencies are absent.

Allowed transitions are determined by $\langle n' | B | n \rangle \neq 0.$

This is the origin of **selection rules**.

Selection rules

The weights of the oscillating terms depend on:

$$\langle \phi_{n',\tau'} | B | \phi_{n,\tau} \rangle .$$

If certain matrix elements vanish, the corresponding frequencies are absent.

Allowed transitions are determined by $\langle n' | B | n \rangle \neq 0.$

This is the origin of **selection rules**.
Bohr frequencies are universal, but their amplitudes depend on the observable.

Quiz

